

WaiYan (Anson) Chan

Ph.D. Candidate / Graduate Research Assistant

University of Wisconsin-Madison

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EDUCATION

University of Wisconsin-Madison

Expected Summer 2028

Ph.D. in Electrical and Computer Engineering

- Minor: Mechanical Engineering
- Thesis: Over-Actuated Axial Flux Permanent Magnet Motor Using PCB Stator
- Advisor: Lei Zhou
- **Novotny Power Engineering Award**

University of Wisconsin-Madison

Sep. 2022 – Jun. 2024

M.S. in Electrical and Computer Engineering

- Thesis: Scaling of High-Speed Surface PM Bearingless Machines
- Advisor: Eric L. Severson

University of Wisconsin-Madison

Sep. 2017 – Dec. 2021

B.S. in Mechanical Engineering

- Project: Compact Low Voltage Induction Machine with 3D Printed Housing
- **Dean's Honor List**

RESEARCH

Precision Mechatronics and Control Lab

Aug. 2024 – Present

EXPERIENCE

Graduate Research Assistant

- Conduct design and control research on an over-actuated, lightweight axial-flux motor, leading the electromagnetic design and fabrication of a PCB-based stator and Halbach-array rotor for high power density rotorcraft/eVTOL applications.

Severson Research Group

Aug. 2022 – Jun. 2024

Graduate Research Assistant

- Additively Manufactured (AM) Stator Housing for High-Speed Bearingless Generator
 - Funding Agency: *U.S. Department of Energy, DE-EE0009138*
 - Designed and fabricated AlSi10Mg AM motor housing with integrated cooling channels for a 100kW, 80kRPM twin bearingless generator for a combined heat and power (CHP) generation system.
- Bearingless Machines (BSPM) for Aerial E-Turbocharger Application
 - Funding Agency: *U.S. Army Research Laboratory, W911NF-20-2-0181*
 - Conducted structural and modal analysis to validate rotor design by identifying critical speed and rotor stress.
 - Modeled and conducted FE analysis of an 8kW 4-DOF twin-stator BSPM, along with component fabrication.
 - Facilitated the development of multi-physics modeling framework for BSPM and created Python scripts for evaluating machine constants and coil inductances in JMAG.

Wisc. Electric Machine & Power Electronic Consortium

Sep. 2019 – Dec. 2021

Undergraduate Research Assistant

- Prototyped a desktop size dynamometer setup capable of characterizing electric machines rated up to 10 N-m of torque and 3000 RPM. Reduced the original setup volume by 35%.

- Collaborated with a team of graduate researchers to prototype a modular terminal box for powering a 200 kW NASA prototype machine. Responsible for sourcing parts and creating CAD models.
- Fabricated BSPM machines by producing stator slot windings and machining components used in the assembly.

TEACHING EXPERIENCE

University of Wisconsin-Madison

Jan. – May. 2026

Teaching Assistant, ME/ECE 577 Automatic Controls Lab

University of Wisconsin-Madison

Sep. – Dec. 2025

Teaching Assistant, ECE 342 Electronic Circuits II

INDUSTRY EXPERIENCE

Motibera, Inc.

Motor Design Engineer Intern

May. - Aug. 2025

- Designed and analyzed a novel bearingless electric motor topology with axial force capability, leading electromagnetic simulation in JMAG and CAD modeling of the prototype, with a patent filing in progress.
- Performed electromagnetic modeling and control oriented analysis of bearingless motor systems, co-authored a conference paper and developed internal tools to support design optimization studies.

Milwaukee Tool

NPD Mechanical Design Engineer

Jan. - Aug. 2022

- Conducted FE analysis on BLDC machines of various sizes and winding configurations; recommended optimal designs for product development using a Pugh Matrix to support informed and cost-effective decision-making.
- Assessed performance requirements of competitor power tools by collecting and analyzing motor thermal characteristics and power output data.

JOURNAL PUBLICATIONS

- J1 T. Noguchi, **W. Chan**, N. Petersen, L. Rapp, and E. Severson, "Opportunities to enhance sCO₂ power cycle turbomachinery with bearingless motor generators," Solar Compass, vol. 12, Dec. 2024, Art. no. 100094 (*DoE Award DE-EE0009138*.)

CONFERENCE PUBLICATIONS

- C6 **W. Chan** and L. Zhou, "Over-Actuated Axial Flux Permanent Magnet Motor using PCB Stator," 2026 IEEE Energy Conversion Conference Congress and Exposition (ECCE), Vancouver, BC, Canada, 2026 (Accepted).
- C5 **W. Chan** and L. Zhou, "Designing for Simplicity: A Novel Segmented-Stator Axial Flux Permanent Magnet Motor with Tape-Wound Cut Cores," 2025 IEEE Energy Conversion Conference Congress and Exposition (ECCE), Philadelphia, PA, USA, 2025, pp. 1-7.
- C4 N. Petersen, **W. Chan**, and A. Khamitov, "Thrust Control via Rotor Skewing in Integrated Twin Bearingless Motor," in 19th International Symposium on Magnetic Bearings (ISMB19), Otsu, Shiga, Japan, Aug. 22, 2025.
- C3 **W. Chan**, T. Noguchi and E. L. Severson, "Scaling of High-Speed Surface PM Bearingless Machines," 2024 IEEE Energy Conversion Congress and Exposition (ECCE), Phoenix, AZ, USA, 2024, pp. 5631-5638 (*US ARL Cooperative Agreement W911NF-20-2-0181*.)
- C2 T. Noguchi, N. Petersen, **W. Chan**, L. Rapp, E. Severson, "Bearingless Motor/Generator Applications in sCO₂ Power Cycles," The 8th International Supercritical CO₂ Power Cycles Symposium, San Antonio, TX, USA, 2024 (*DoE Award DE-EE0009138*.)

	C1	T. S. Slininger, W. Chan , E. L. Severson and B. Jawdat, “An Overview on Passive Magnetic Bearings,” 2021 IEEE International Electric Machines & Drives Conference (IEMDC), Hartford, CT, USA, 2021, pp. 1-8 (<i>NSF Grant 1938610</i> .)	
INVENTION DISCLOSURES	I2	L. Zhou, W. Chan , “Rim Driven Rotorcraft Motor with Blade Stabilization” Co-inventor, U.S. Patent Application No. 19/745,999, filed Jul. 17, 2026.	
	I1	N. Petersen, W. Chan , and A. Khamitov, “Integrated Bearingless Motor with Axial and Radial Magnetic Levitation,” Co-inventor, PCT Application No. PCT/US26/41085, filed August 13, 2026 (claiming priority to U.S. Provisional Application No. 63/864,569).	
ORAL PRESENTATIONS	O1	<i>Scaling of High-Speed Surface PM Bearingless Machines</i> , IEEE Energy Conversion Congress and Exposition (ECCE), Phoenix, AZ, USA, 2024.	
POSTER PRESENTATIONS	P1	<i>Designing for Simplicity: A Novel Segmented Stator Axial Flux Permanent Magnet Motor with Tape Wound Cut Cores</i> , IEEE Energy Conversion Congress and Exposition (ECCE), Philadelphia, PA, USA, 2025.	
AWARD AND HONORS		Novotny Power Engineering Award College of Engineering, University of Wisconsin-Madison, Madison, WI	2026
		Best Presentation Award 19th International Symposium on Magnetic Bearings (ISMB19), Otsu, Japan	2025
SKILLS		Hands-on: Electric Machine Fabrication, Component Machining and Assembly, CAD Design and Drafting, GD&T Tools: Python, Git, SolidWorks, NX, ANSYS, MATLAB / Simulink, Altium, LabView, MAGNET, FEMM, JMAG, $\text{\LaTeX}$	
CERTIFICATIONS		Fundamentals of GD&T ASME Y14.5, Applied Geometrics, Inc.	Apr 2022